BIOL-1 Practice Test 03 — Answer Sheet

Exam 03 Preparation (Modules 12–16)

Name:		Date:		Score:	

✂️ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (25 questions) and Part B: Fill in the Blank (10 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (25 questions)

1.		**10.**		**19.**	
2.		**11.**		**20.**	
3.		**12.**		**21.**	
4.		**13.**		**22.**	
5.		**14.**		**23.**	
6.		**15.**		**24.**	
7.		**16.**		**25.**	
8.		**17.**			
9.		**18.**			

Part B: Fill in the Blank (10 questions)

1.		**6.**	
2.		**7.**	
3.		**8.**	
4.		**9.**	
5.		**10.**	

BIOL-1 Practice Test 03 — Part C: Short Answer (4 questions)

Free response. Write clearly in the lined spaces.

1. **(Module 12)** In two or three sentences, define **evolutionary fitness** as used in this course, and name **one** line of **evidence** for common descent (fossils, comparative anatomy, biogeography, or molecular) with a brief what-it-shows.
2. **(Module 13)** Distinguish **bottleneck** and **founder** effects. Then name the three **modes of natural selection** on a continuous trait (directional, stabilizing, disruptive) in one line each.
3. **(Module 14)** Give one **prezygotic** and one **postzygotic** example of a **reproductive barrier** (one sentence each). In one sentence, contrast **allopatric** and **sympatric** speciation.
4. **(Module 15)** Contrast **density-dependent** and **density-independent** limiting factors with one **example of each** (one sentence on each type). In two sentences, use the **~10%** energy rule to explain **why** food chains with many links support **fewer** apex consumers than a short chain from the same primary production.
5. **(Module 16)** Build a short systems explanation: choose one biological scenario from this unit and connect **one evolutionary idea**, **one population/ecology idea**, and **one piece of evidence** into a single cause-and-effect explanation.

Response 1 — Question Number: ______

Response 2 — Question Number: ______

Response 3 — Question Number: ______

Response 4 — Question Number: ______

Response 5 — Question Number: ______

BIOL-1 Practice Test 03 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 12: Darwin and Evolution

1. In the logic of natural selection, which observation states that more offspring are produced than can usually survive to reproduce?
    - A) Inheritance of acquired characters
    - B) Differential reproductive success
    - C) Overproduction of offspring
    - D) Stabilizing selection only
2. A trait increases **evolutionary fitness** in a given environment if individuals with that trait tend to:
    - A) Contribute more alleles to the next generation than competitors do
    - B) Acclimate in one lifetime to match the parent phenotype
    - C) Look larger than their competitors
    - D) Survive long enough to be counted, even with zero offspring
3. Which statement is **most** consistent with **descent with modification**?
    - A) Each species was created in its final form and does not change
    - B) Lineages change over time; related species share ancestors with modification
    - C) Only fossils evolve, not living species
    - D) The strongest individual in a species can change the population's heritable gene pool in one lifetime, without reproduction

4. Homologous forelimb bones in a human, a cat, and a whale are used as evidence for evolution because they:
    - A) Suggest a shared ancestry with structural change for different uses
    - B) Disprove the existence of a fossil record
    - C) Prove that whales evolved from modern humans
    - D) Show identical functions in all three species
5. A population of moths in a clean forest is mostly light-colored; in a sooty industrial region the population becomes mostly dark. The clearest **selection** label for that shift, if the environment favors the dark extreme, is:
    - A) Stabilizing selection
    - B) The founder effect
    - C) Directional selection
    - D) Disruptive selection
6. Which option best states why **populations** evolve, not **individuals**?
    - A) One hungry bird cannot affect allele frequencies at all (bird effects are not evolution)
    - B) Natural selection has nothing to do with heritable variation
    - C) Evolution is a change in allele **frequencies in a population** across generations
    - D) Individuals can change their DNA by lifting weights

Module 13: How Populations Evolve

1. The definition of **microevolution** used in this course is:
    - A) A **change in allele frequencies** in a population over time
    - B) A change in an individual’s size during one lifetime
    - C) A synonym for any mutation in a body cell only
    - D) The origin of a new phylum in one generation

2. A few finches from the mainland start a new island population. The island gene pool reflects only the alleles the founders brought. This is:
   - A) Stabilizing selection on beak size
   - B) The bottleneck effect
   - C) The **founder** effect (genetic drift)
   - D) Convergent evolution
3. A volcanic eruption kills 95% of a squirrel population at random, then survivors repopulate. The loss of many alleles from the pre-eruption pool is:
   - A) A founder effect in a new island colonization from another continent only
   - B) A **bottleneck** (genetic drift)
   - C) A proof that survivors were the "most fit" for every possible future environment
   - D) Interspecific competition in the absence of intraspecific variation
4. In **stabilizing** selection on a trait (for example, birth weight in some human data sets):
   - A) Allele frequencies must stay exactly at Hardy–Weinberg equilibrium forever
   - B) The population's average always moves to one tail of the range
   - C) The **intermediate** phenotype is favored; extremes are selected against
   - D) Both extreme phenotypes are favored
5. The Hardy–Weinberg equilibrium model is useful mainly because it:
   - A) Describes a real population that never evolves
   - B) Serves as a **mathematical baseline** to measure evolutionary change in real data
   - C) Guarantees $p^2 + 2pq + q^2 = 1$ in every generation without assumptions
   - D) Replaces the biological species concept
6. The movement of alleles into or out of a population through **migration and mating** is:
   - A) Genetic drift in an infinitely large population
   - B) **Gene flow**
   - C) A postzygotic barrier to reproduction

◦ D) Non-random mating only

Module 14: Macroevolution

1. Under the **biological species concept**, two groups are different species if they:

    ◦ A) Are **reproductively isolated** in nature (they do not interbreed to produce viable, fertile offspring **between** the two groups, while within each group interbreeding can occur)
    ◦ B) Have different number of cells
    ◦ C) Live on different trees in the same forest, without any other test
    ◦ D) Look different in a field guide, regardless of interbreeding

2. Fireflies of two species use different **flash** patterns, so they rarely attempt mating. This isolating mechanism is best classified as:

    ◦ A) The bottleneck effect
    ◦ B) Postzygotic reduced hybrid fertility
    ◦ C) A mechanical postzygotic barrier
    ◦ D) A **prezygotic behavioral** barrier

3. A mule, the hybrid of a horse and a donkey, is typically sterile. This is an example of:

    ◦ A) A prezygotic gametic barrier
    ◦ B) The founder effect
    ◦ C) A **postzygotic** barrier to gene flow (reduced hybrid fertility)
    ◦ D) Stabilizing selection in horses

4. A new mountain range **physically splits** a population; over long time, the two lineages no longer interbreed. The classic geographic mode of speciation is:

    ◦ A) A requirement that drift never occurs
    ◦ B) **Allopatric** speciation
    ◦ C) Sympatric speciation in one meadow
    ◦ D) The bottleneck effect in both lineages on purpose

5. A plant can form a new, reproductively isolated **polyploid** lineage in a few generations, often in the same area as a parent type. This route is a frequent example of:

    ◦ A) The bottleneck effect only
    ◦ B) Only macroevolution in animals, not plants
    ◦ C) A prezygotic postzygotic hybrid
    ◦ D) **Sympatric** speciation (e.g. via polyploidy)

6. The statement "macroevolution is different chemistry from microevolution" is **false** in this course because:

    ◦ A) The fossil record is irrelevant to modern biology
    ◦ B) All macroevolution happens in a single meiosis
    ◦ C) No DNA is involved in speciation
    ◦ D) The same **population**-level processes (selection, drift, flow, etc.) can accumulate to reproductive isolation and larger-scale diversity

Module 15: Population and Systems Ecology

1. A population introduced to a new habitat with abundant resources for a time shows growth that **accelerates** (as long as the ceiling is not yet reached). That early phase is classic for:

    ◦ A) An **exponential (J**-shaped) phase of growth
    ◦ B) A carrying capacity of zero by definition
    ◦ C) A flat line at carrying capacity from year one only
    ◦ D) Only density-independent regulation, never K

2. A population levels off as resources become scarce, approaching a relatively stable size. That ceiling is:

    ◦ A) A genetic bottleneck
    ◦ B) **Carrying capacity (K)** in a logistic (S) model
    ◦ C) A proof that 100% of energy moves to the next level
    ◦ D) A synonym for the founder effect

3. A fungal disease that spreads more easily in **dense** host stands is an example of a factor often:

    ◦ A) Only density-independent like wind
    ◦ B) Unrelated to per-capita rates
    ◦ C) Never biological
    ◦ D) **Density-dependent** in its impact on the population

4. A severe freeze kills a **similar fraction** of a plant population across sparse and dense patches (the freeze does not "care" about crowding the way a contagion might). Treated as a **limiting** event in textbook classification, the freeze is **most** like:

    ◦ A) A **density-independent** disturbance (still harmful to the population)
    ◦ B) A biogeochemical cycle for nitrogen only
    ◦ C) Only density-dependent competition for light
    ◦ D) A trophic level

5. The **~10% rule** between trophic levels implies that, compared to eating **producers** directly, a diet of **higher** trophic levels (many steps) to deliver the same human calories will tend to need:

    ◦ A) Less primary production because carnivores are 100% efficient
    ◦ B) The same amount of energy at every level, by definition of food chains
    ◦ C) Decomposers to add energy from the sun to dead matter
    ◦ D) **More** total primary production at the base of the food web

6. In an ecosystem, **chemical elements** (e.g. carbon, nitrogen in nutrients) are **recycled**, while the energy fixed by **photosynthesis** is eventually:

    ◦ A) **Lost** largely as **heat** along the way; not recycled the way a carbon atom can be
    ◦ B) Transferred 100% from producer to all consumers combined
    ◦ C) Only stored in rocks, not in life
    ◦ D) Re-used indefinitely by the same photon inside every cell with no loss

7. Fungi and bacteria in soil return nutrients from dead material to **forms** plants can use again. In contrast to energy, this recycling pattern supports the idea that:
    - A) Decomposers are always apex predators in every web
    - B) Energy cycles exactly like water in the stratosphere only
    - C) Producers do not use soil nutrients
    - D) Decomposers are central to **nutrient** cycling for producers, while **energy** flows in one direction through the system

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Part B: Fill in the Blank (10 questions)

Word Bank: Allopatric, allele frequencies, decomposers, density-dependent, gene flow, homologous, K, prezygotic, stabilizing selection, sympatric

1. A population’s long-term, stable ceiling in a logistic (S) model is the carrying capacity, symbol _______________.
2. The formal focus of **microevolution** is a change in ___________________ over generations in a population.
3. A barrier that works **before** a zygote—such as different breeding times—is a ___________________ barrier to reproduction (category name).
4. A limiting factor (such as a contagious disease) whose impact on survival or reproduction can intensify with crowding is often ___________________.
5. The movement of alleles into or out of a population through migration and mating is called ___________________.
6. The forelimb bone pattern shared by many tetrapods with different functions (human arm, bat wing, whale flipper) is ___________________ structure (in comparative anatomy, from common ancestry with modification).
7. Fungi and many bacteria, which break down dead material and return elements to the soil, are ___________________ in nutrient cycling (category name).

8. Modes of natural selection: when the **middle** of a trait's range is favored and the extremes are selected against, the pattern is often __________________.
9. A river splits a continuous population; the two sides evolve in isolation. That geographic mode of speciation is often __________________ speciation.
10. A new species arises **without** geographic separation, as in some plant **polyploidy** lineages, is a textbook example of __________________ speciation.